

DOMAIN: AGRI / WATER / CLIMATE - TECH



WATERFALL

AUTOMATION PRIVATE LIMITED

YOUR FARM
IN YOUR PALM

Jasveer Singh (IIT Bombay Alumni)
Founder & CEO



IoT-based, AI-ready smart irrigation
MAKING IRRIGATION MANAGEMENT RELIABLE, EFFICIENT & SCALABLE



Farmers
200+



Land under SICCA
1000+
Acres



ARR
1.2Cr+
Rupees



Water saved
1B+
Litres



Electricity saved
50K+
Units



CO₂ Saved
50+
Tons

Current Irrigation Practices are Inefficient

! Inconvenience



Inconvenient electricity schedules
are at night times

Wild animal threat -
deaths due to snake bites and other
animal attacks are common

Electrocution Hazard

Poor accessibility -
no proper roads, street lights

Affected secondary occupation -
due to labour shortages and night-time
schedules, resulting in reduced
livelihoods



Low Profitability

Over irrigation -

- Yield loss due to root rot
- Fertilizer wastage
- Water wastage
- Soil degradation

Under irrigation -

- Yield loss due to stress

Labour inefficiency -

- Labour may address convenience
but fail to deliver precision



Existing Solutions Are Not भारत Farmer Friendly

Failing 13 Cr Farm Landholders

EXISTING AUTOMATION TECHNOLOGIES HAVE PENETRATED ONLY 0.1% OF THE INDIAN MARKET IN LAST 25 YEARS:

- ❗ Expensive/unaffordable
- ❗ Not farmer friendly
On-site presence required to set up irrigation schedules
Difficult to operate as it needs technical skills
- ❗ High overhead cost and effort as it required to
Change entire pipeline structure to centralized
Construct a room to accommodate automation system
- ❗ Unreliable soil sensors
Slow response rate
- ❗ Maintenance support too expensive
Requires a skilled personnel and higher downtime.

A representative image of a bulky, expensive controller needing separate room structure to manage centralized pipeline at farm adds to overhead cost





USP



Mobile Friendly

- Remotely accessible
- Intuitive and user-friendly,
- suitable for farmers with minimal technical knowledge



Retrofit Design

- Saves lakhs of overhead costs; no change in pipeline structure required .
- No extra room space required; fits on a wall mount



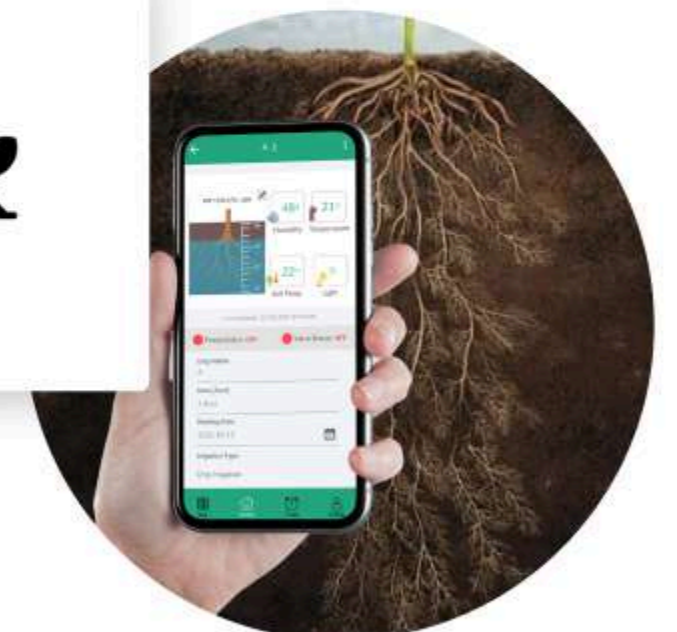
Easy Installation & Low Maintainance

- Plug & play (DIY)
- Easy support
- Only 24 hrs. downtime



Precise Irrigation

- Quick response, Root-zone soil moisture sensor
- 50% more water savings

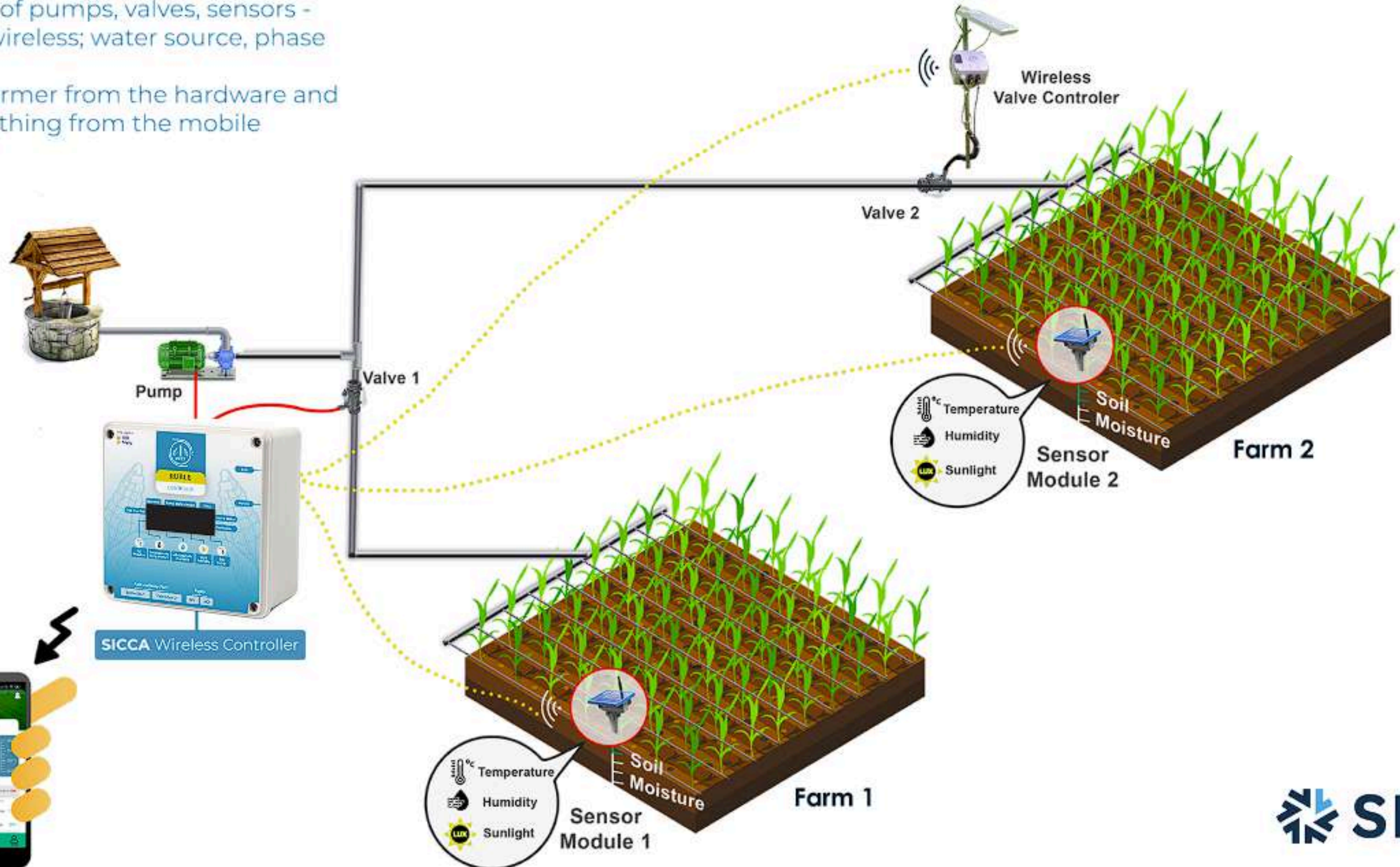


5x Affordable

SICCA-powered farm - Wireless Sensor, Valve & Pump nodes

Solution is very flexible;

- Can control n no. of pumps, valves, sensors - wired as well as wireless; water source, phase does not matter.
- It abstracts the farmer from the hardware and can control everything from the mobile

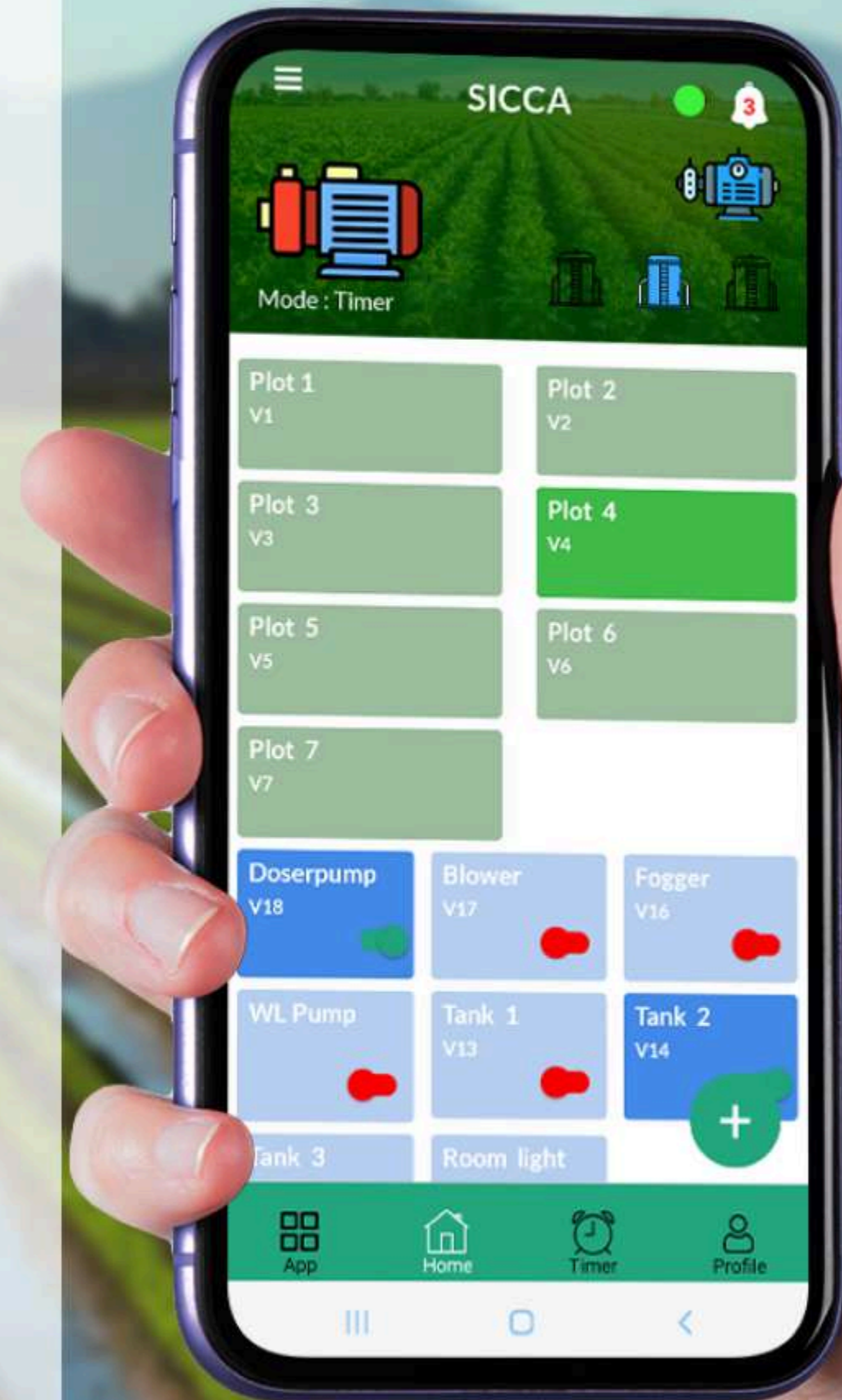
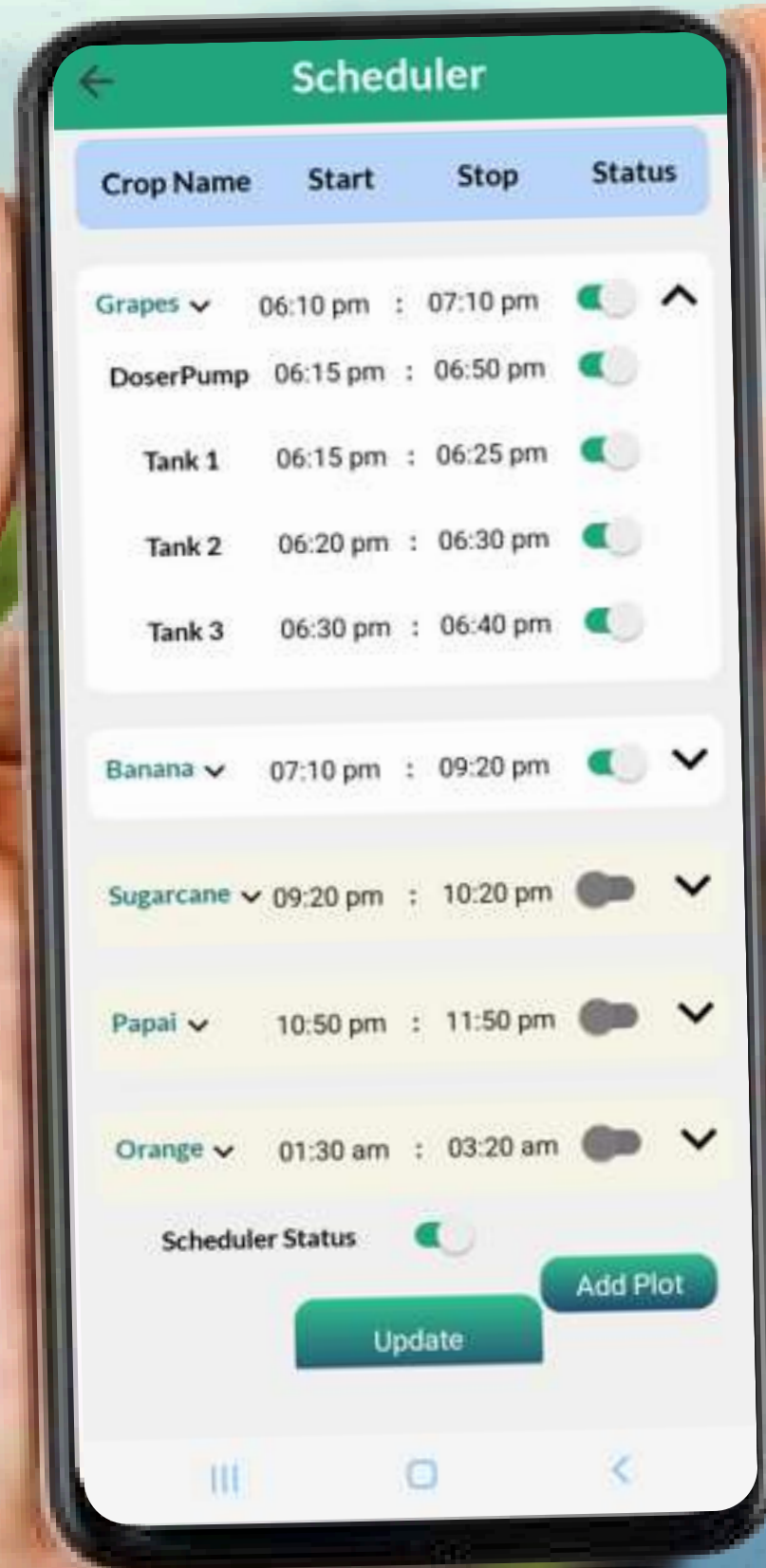


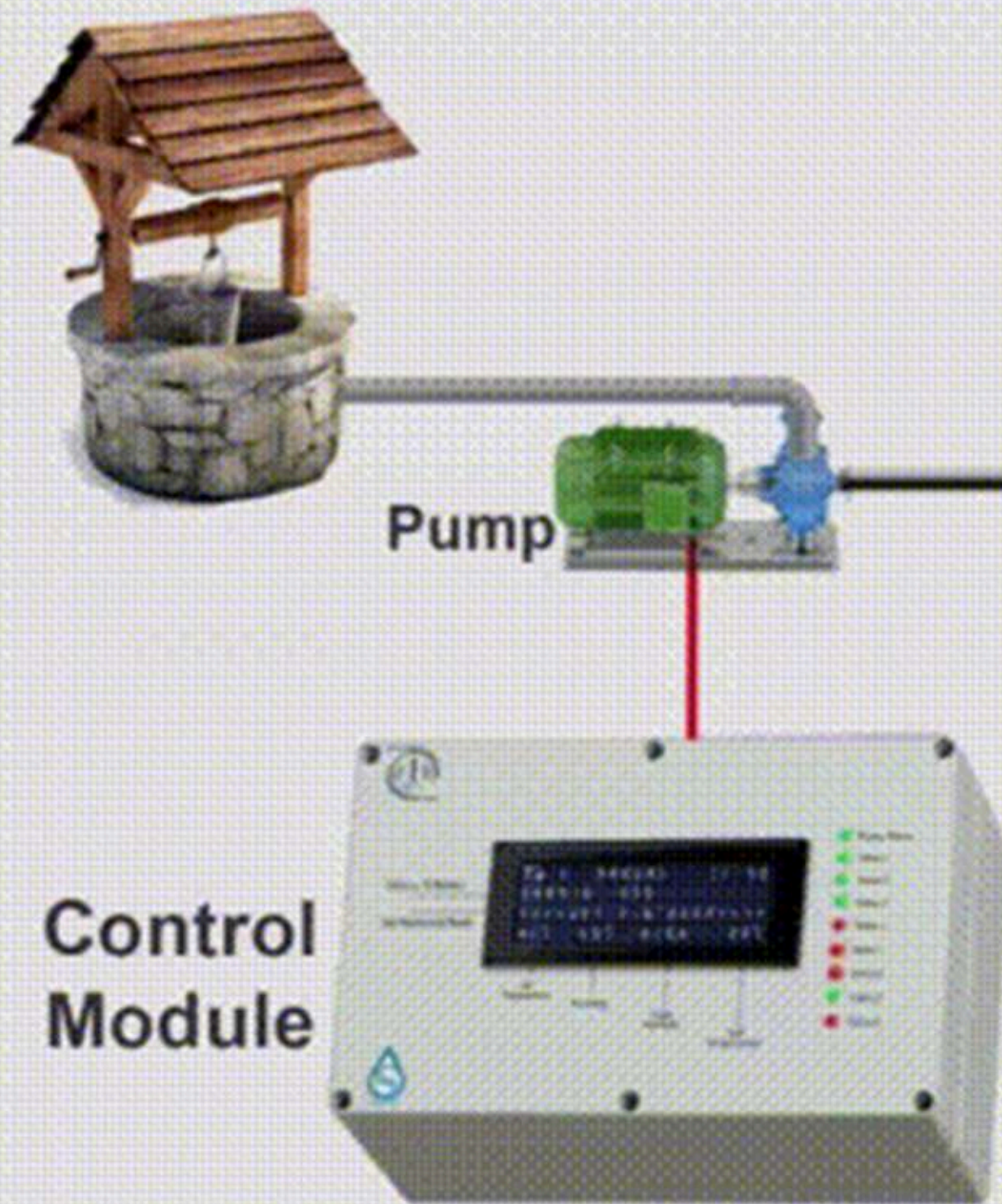


**FULL CONTROL OF FARM
THROUGH MOBILE**

**CAN SCHEDULE FERTIGATION
AND WATER IRRIGATION**

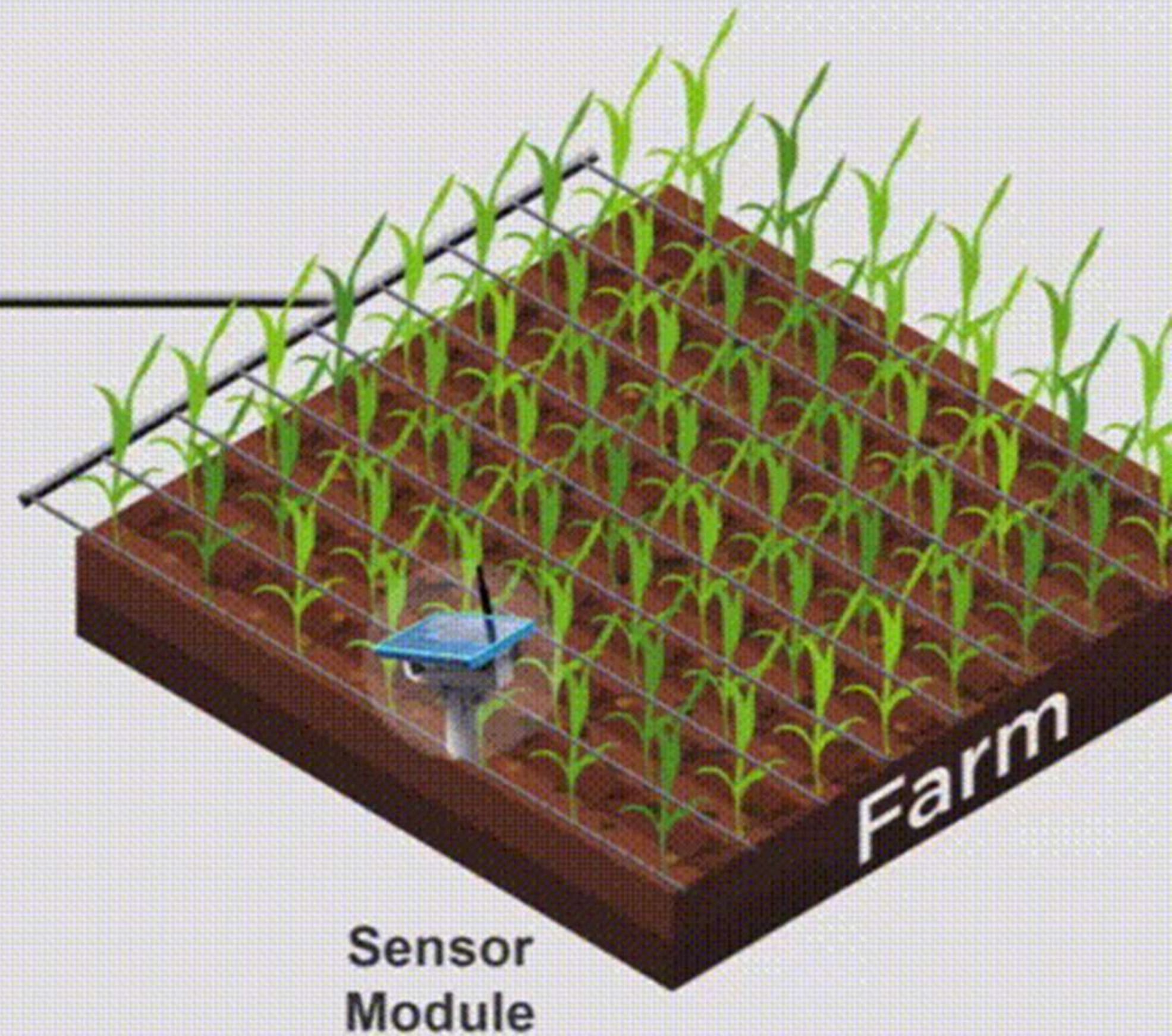
ELECTRICITY TURBULENCE





**Control
Module**

This slide was meant to have an animation,
feel free to watch it online by clicking this link



**Sensor
Module**

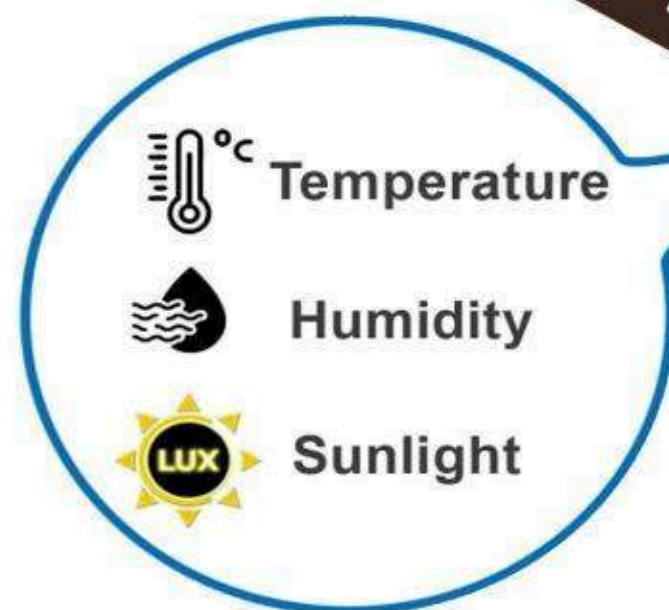
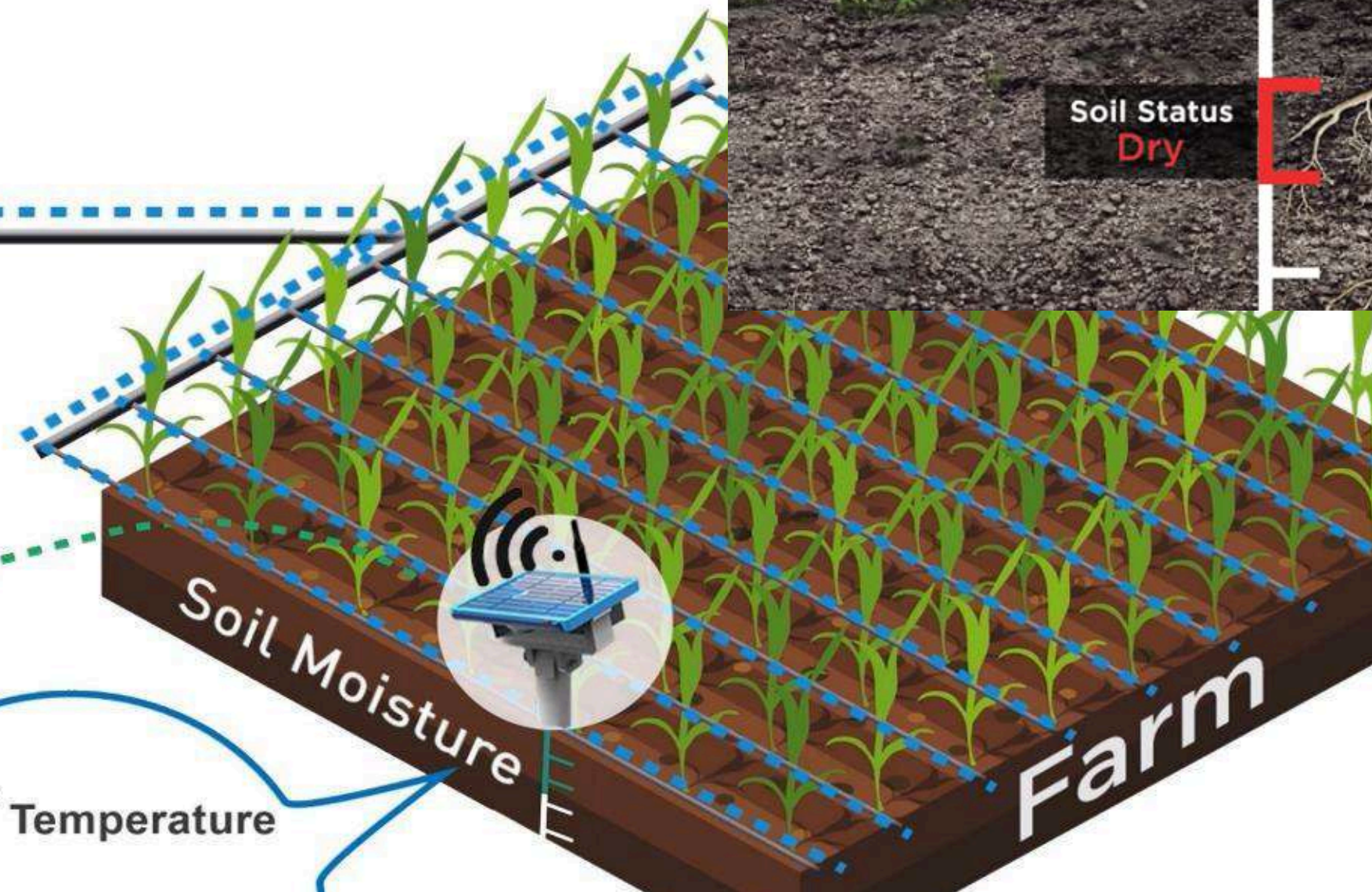


Pump



On

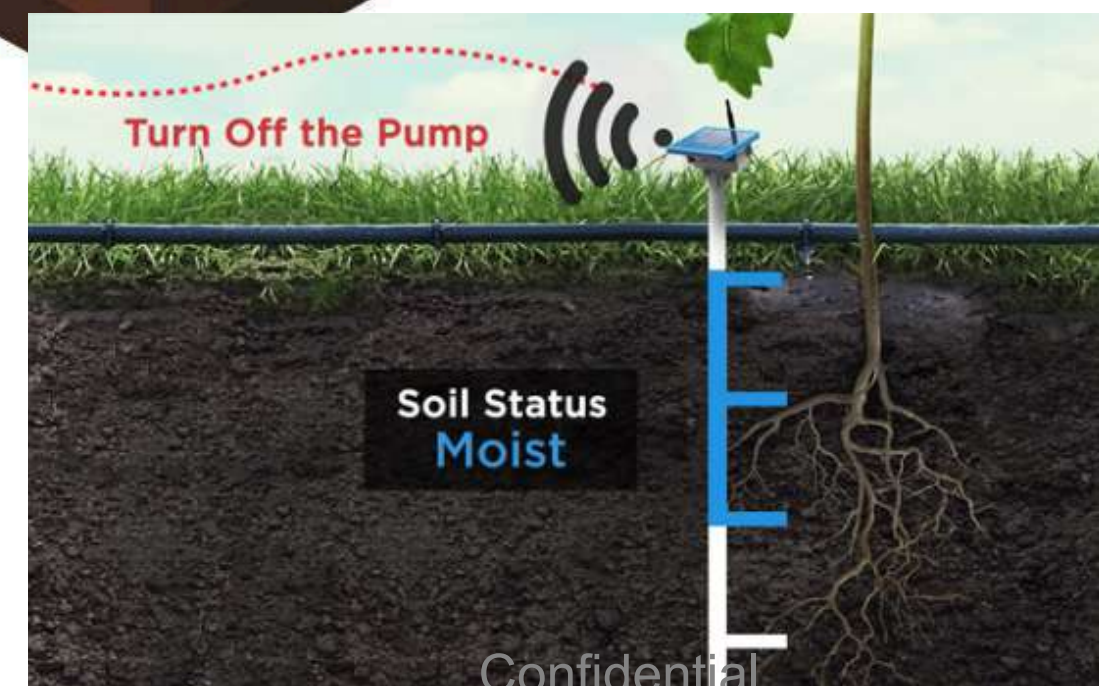
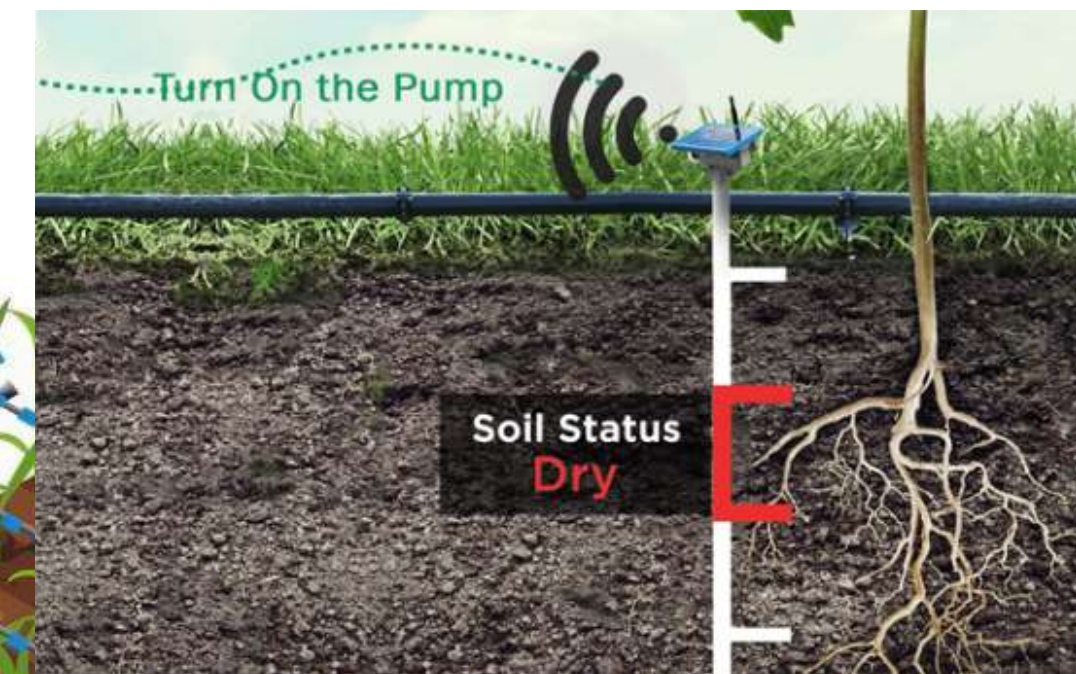
Control
Module



Sensor
Module



Mobile
App

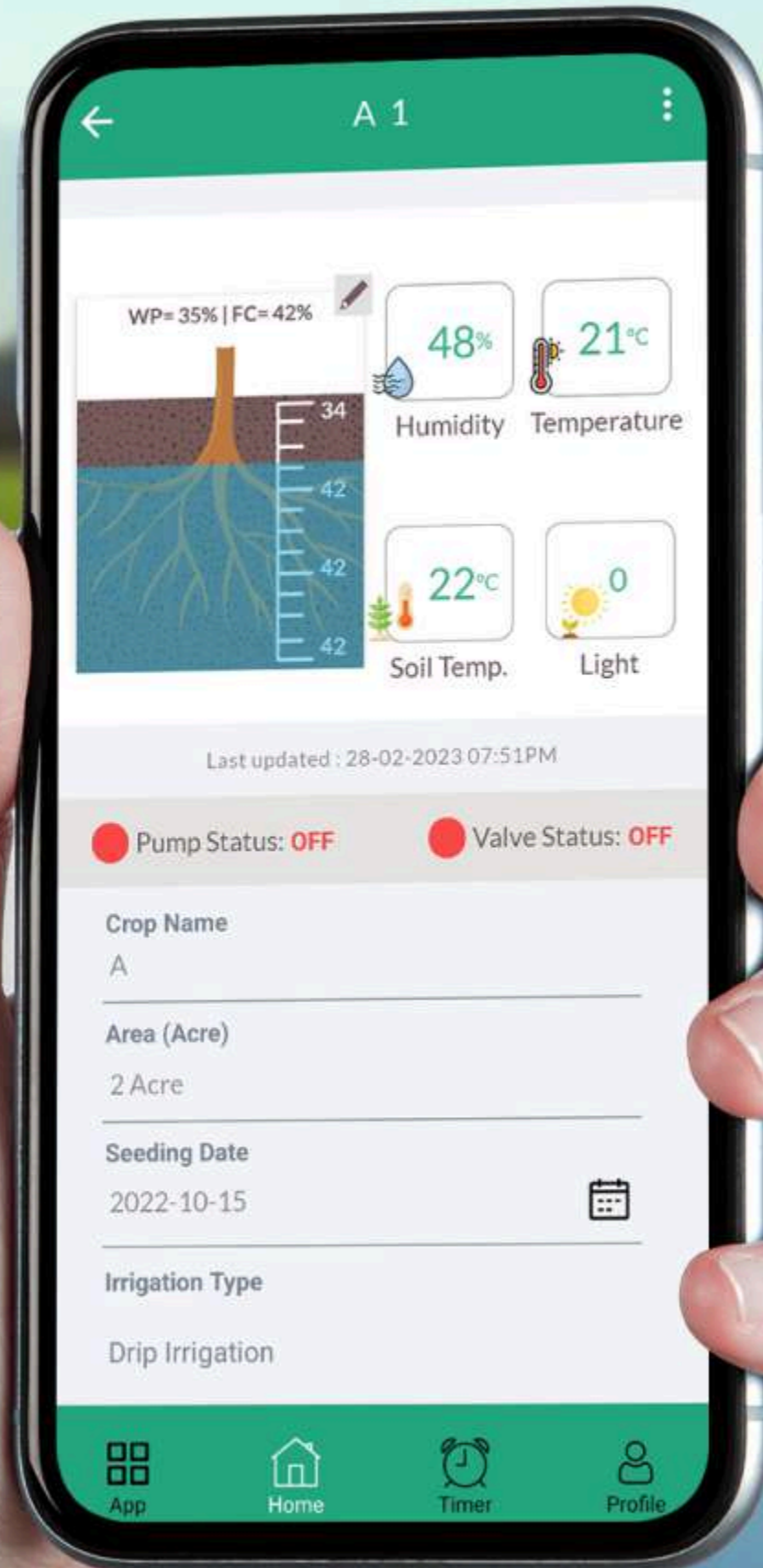


Confidential



mySICCA Mobile app

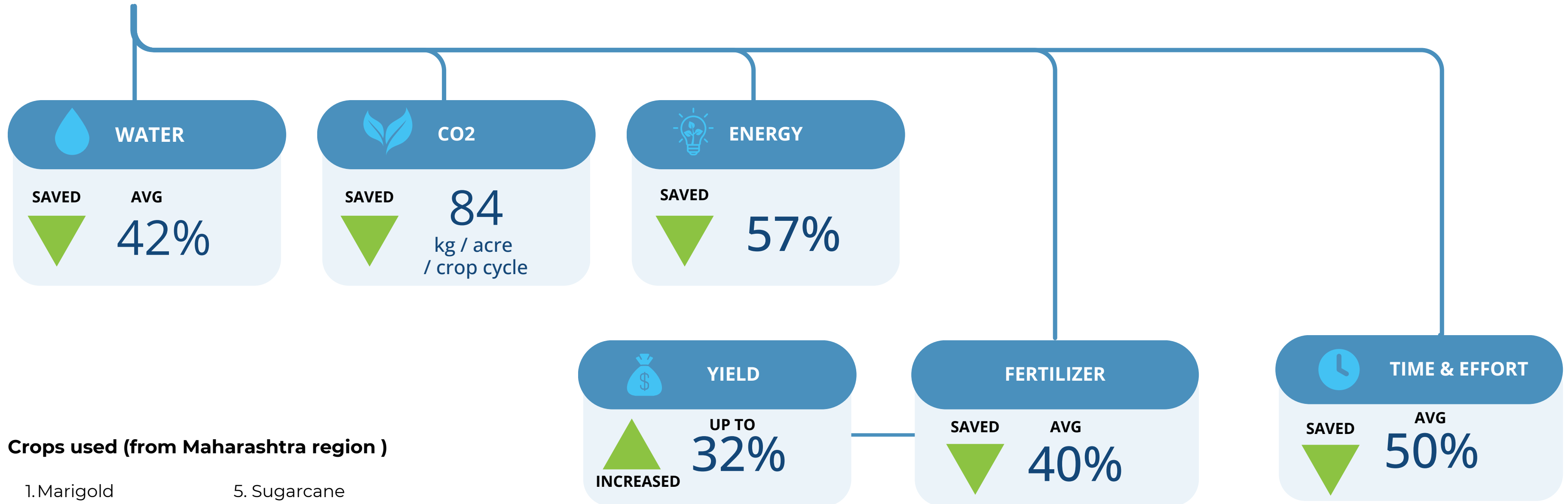
- Precise irrigation as per crop need
- Sensor-based automation
- Quick-response sensors
- Save 50% more water





BENEFITS

PILOT REPORT SUMMARY – DEC. 2022 UNIDO – CII'S FACILITY FOR LOW CARBON TECHNOLOGY DEPLOYMENT



Crops used (from Maharashtra region)

1. Marigold
2. Capsicum
3. Drumstick
4. Pomegranate
5. Sugarcane
6. Grape
7. Geranium

Core Team



Jasveer Singh

Founder & CEO

- M.Tech (Electronics), IIT Bombay
- Serial Entrepreneur, 7 yrs - startups & AgriTech space
- Exp. in working with agri-communities to uplift livelihood in drought-affected regions



Gajanan Shidardi

Co-Founder & CMO

- M.Tech (Elex), SGGS Nanded
- Farmer – Special crops
- Exp. in rural agri-marketing
- Well known in MH Farmer network



Faizan Shaikh

Co-Founder & CTO

- Serial AgriTech Entrepreneur
- Alumni - Tata DISQ, Nashik & Stanford Seed-Spark Program



Dr. Manmeet Singh

Consultant (AI Expert)

- Scientist (Climate Change), IITM
- PhD – Climate Studies, IIT Bombay
- Fulbright Fellow, UT Austin
- Expertise in GIS (satellite data processing)

Advisors



Mayank Gandhi

Managing Trustee at Global Vikas Trust

- Founder, Global Vikas Trust
- Increased farming income by 5 times
- Fal-Nayak & Jal-Nayak of drought ridden Marathwada region
- Built state of the art Krishikul, farmer training center in Parli, MH

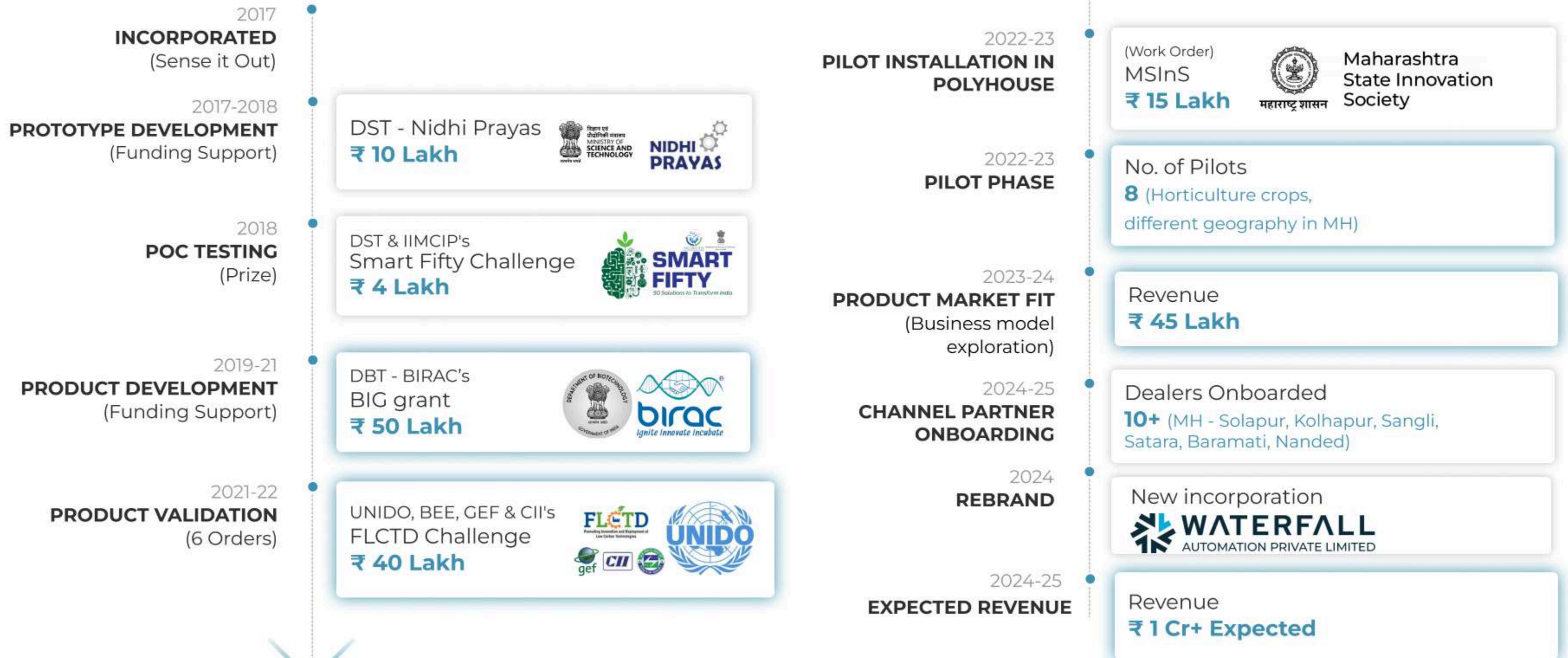


Collin Downey

Founder | Zinc Tanks

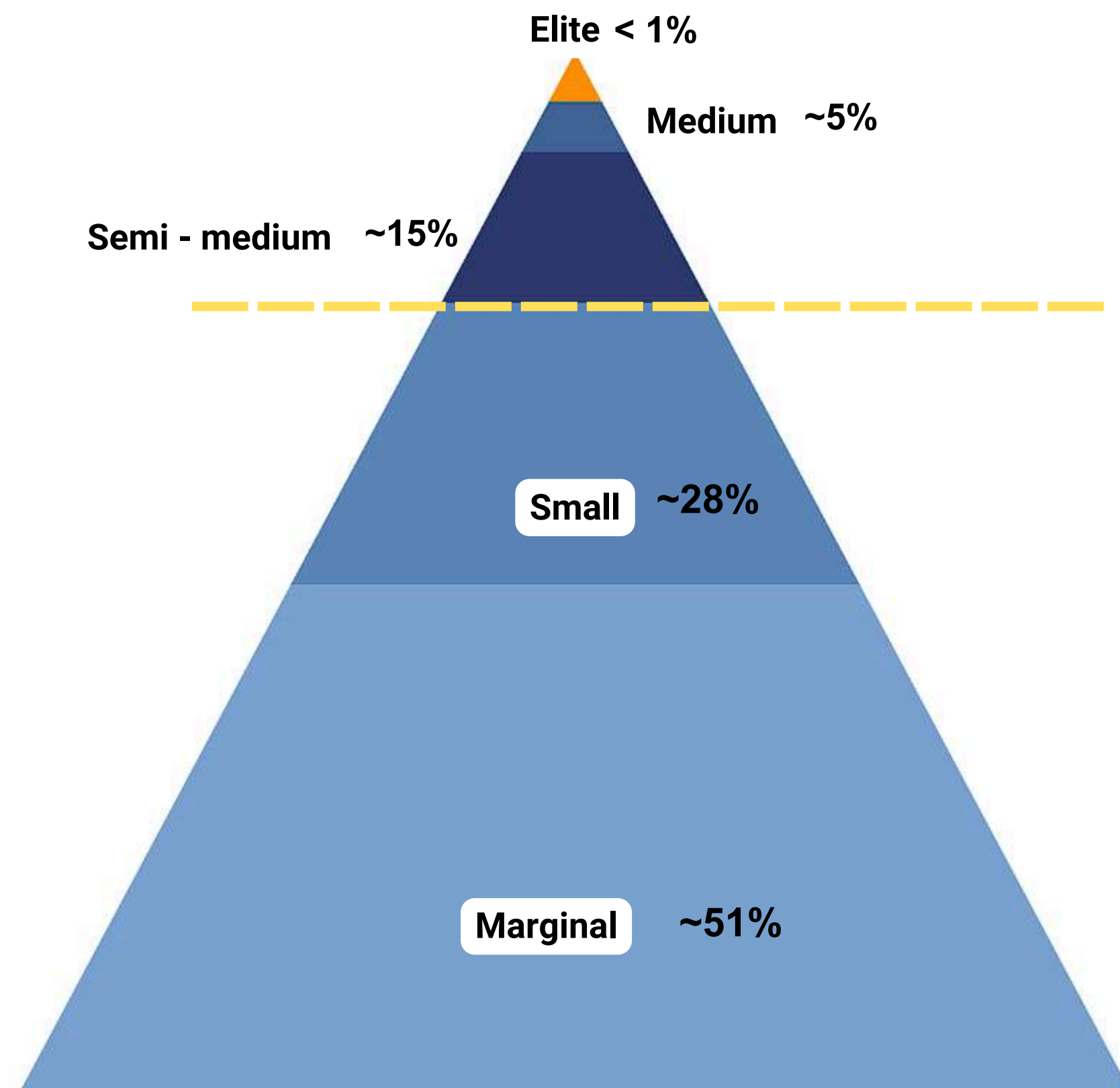
- 35+ years of experience in B2B Agri sales
- Experience in international sales and market expansion, particularly in emerging markets.
- Built strong relationships with distributors, dealers, and agri-businesses across key markets.

Journey So Far



Product technically validated by United Nations Industrial Development Organization (UNIDO) & CII

CUSTOMER SEGMENT

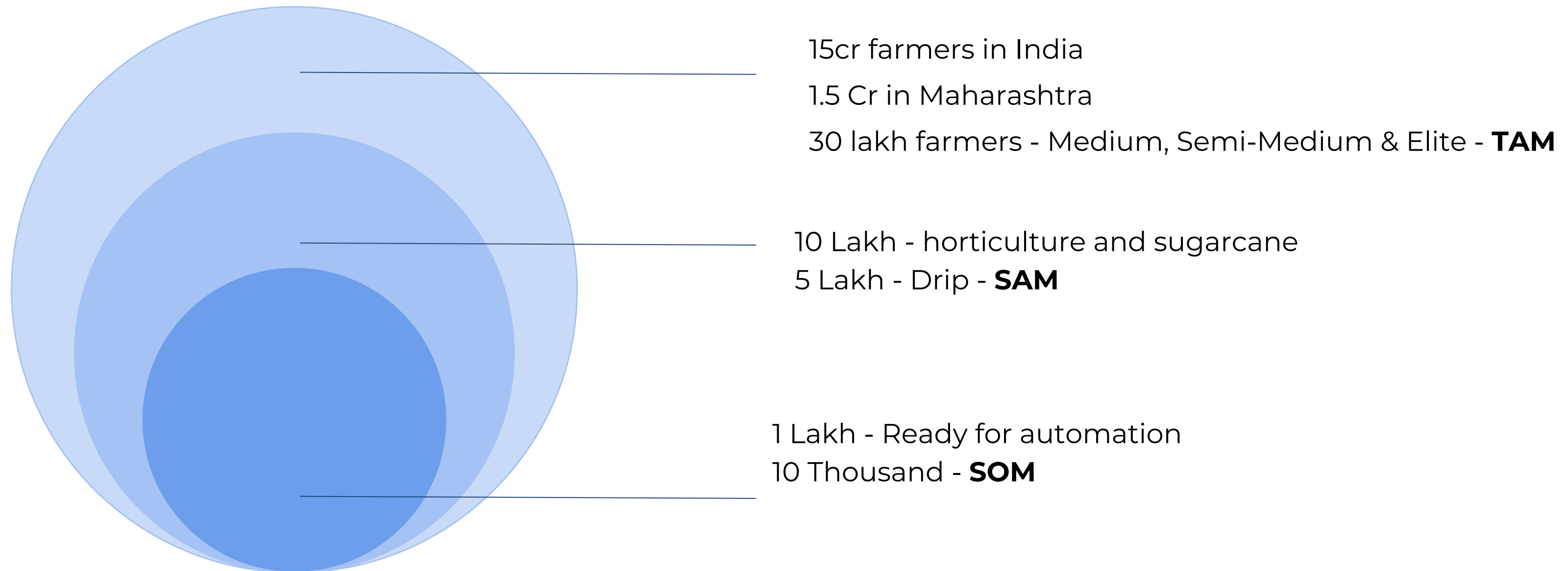


Farmer Classification by Land Holdings in MH

WHY LARGER FARMS?

- Early adopters
- Crops grown: Horticulture (fruits, vegetables, flowers, spices, medicinal crops) and sugarcane.
- High adoption of drip irrigation for precision farming.

MAHARASHTRA'S AGRICULTURAL LANDSCAPE:



With a ticket size of ₹2 lakhs, by catering to just 10,000 farmers, we can generate ₹200 crores in revenue from Maharashtra alone.

Applying the same strategy to Gujarat, Andhra Pradesh, Telangana, Kerala, Karnataka, and Tamil Nadu, we could achieve a revenue of ₹1,000 crores across these states.

COST BENEFIT



Meet Ankush, a hardworking farmer from Maharashtra with a 10-acre sugarcane farm. For years, he struggled with traditional irrigation—let’s say it wasn’t exactly “watering” down his problems. Then Hari found SICCA and gave it a shot. Let’s just say his crops are now thriving, unlike his old irrigation was!

Farm Size - 10 acres	Without SICCA	With SICCA	Benefits
Average Yeild (per acre)	40 tons	60 tons	20 tons
Selling Price (per ton)	₹3,000	₹3,000	₹3,000
Total Revenue (INR)	₹12,00,000	₹18,00,000	₹ 6,00,000

- Benefits that can be additionally factored:
- Labor savings: Less manual work required for irrigation and fertilization.
 - Fertilizer savings: Precise delivery reduces wastage.
 - Water conservation: Efficient irrigation ensures optimal use.



PRODUCT PRICING

Gross margin : 47%

5G IoT Devices	Crops	One Time Hardware Cost	Annual Subscription (No dealer cut)
SICCA One (Labour problems & inaccuracy because of labour)	Sugarcane	₹ 2,00,000	₹ 6,000
SICCA Smart (Precise application of water & fertilizers using soil sensors)	Horticulture	₹ 3,50,000	₹ 9,000
SICCA Advanced (Precision for Water, Fertilizers & Climate control)	Polyhouses	₹ 7,00,000	₹ 15,000

(1 yr warranty & 1 yr subscription included, AMC (optional) @15% from 2nd year)

FUTURE PRODUCT

Mission Wapsa Wireless Sensor Node	₹ 20,000 (<2.5 acre)	₹ 2,000
--	-------------------------	---------

Channel

- Dealer-Distributor network of drip irrigation companies in rural areas. They have a credit facility & trust-based approach locally.
- They have the following roles & responsibilities:

1. Market access / Lead generation
2. Ensuring timely payments by farmers
3. Installation
4. Maintenance

Progress in Channel Partnerships:

- Onboarded Partners:
 - 10 dealers already onboarded.
- Ongoing Engagements:
 - 15 dealers currently in discussions across Maharashtra.
 - 2 distributors .
 - Multiple B2B approaches actively in progress.
- Revenue Sharing:
 - Dealer Margin: 15%.
 - Distributor Margin: 15%



ROADMAP AHEAD

2025-26

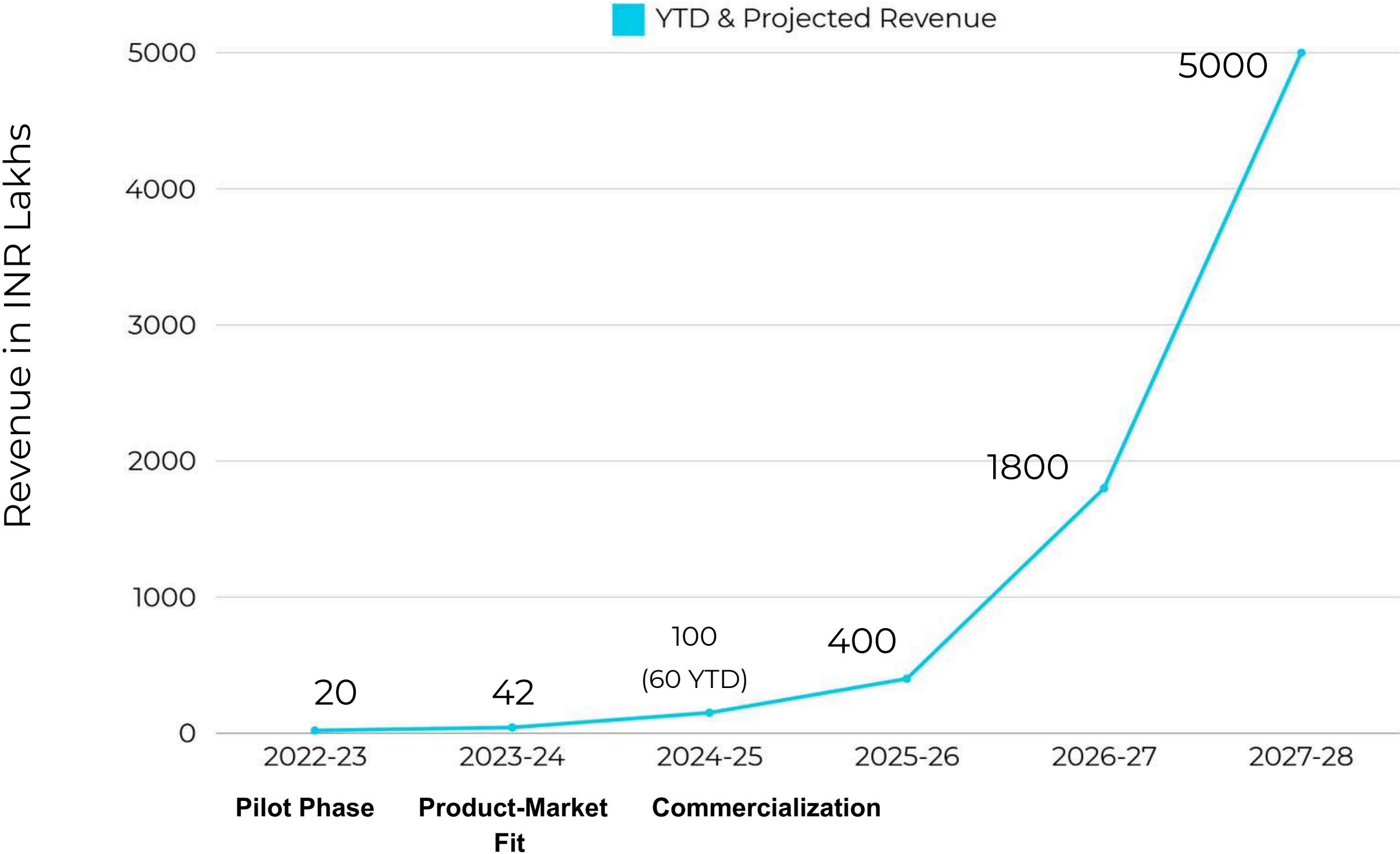
Scale in Maharashtra & touchpoints in neighbouring states

2026-27

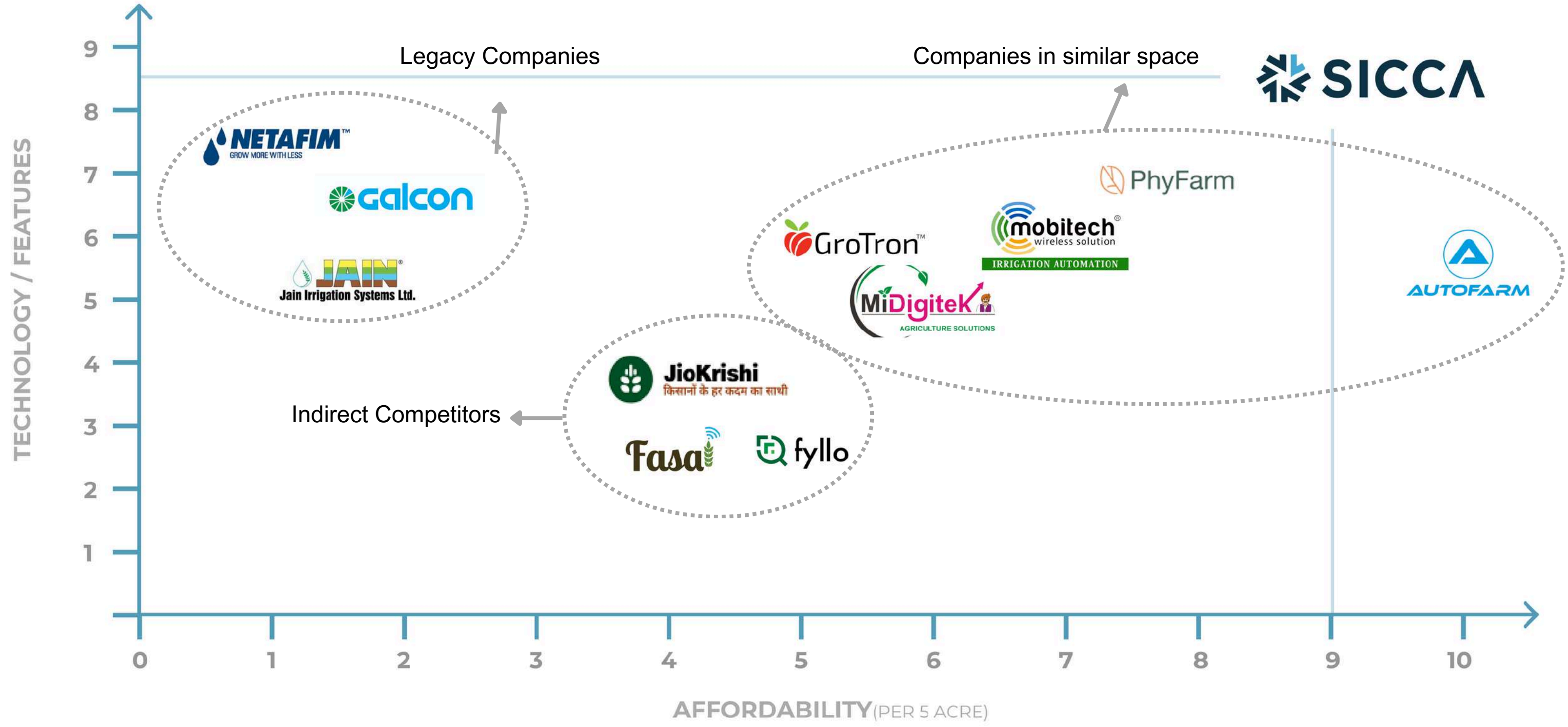
Scale in 5 Indian states and touchpoints in East African, MENA, SEA Countries

2027-28 & 2028-29

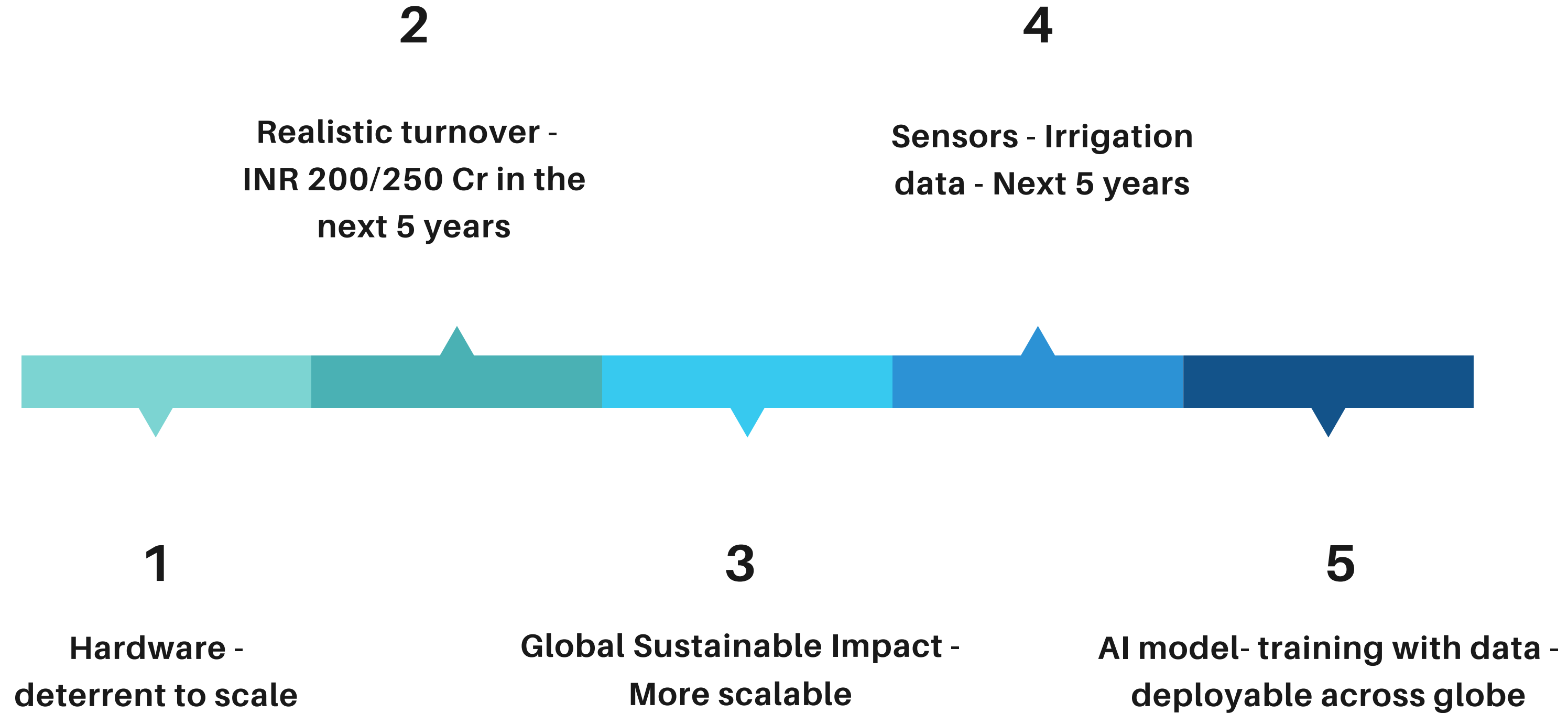
Deepening reach in selected markets and broadening offerings globally



Competitor Landscape in India



AI PRODUCT ROADMAP



Funds Required

₹ 1.75 Cr

At a **pre-money**
valuation of ₹ 20 Cr

Use of Funds



Pacing up sales & Installations
Feet on ground



Robust backend customer
support for scale



For marketing to boost
adoption rates



For boosting production cycle

Testimonials

SICCA Smart Farmers



Dr. Ankush Chormule 15 Acre

SICCA Sensors helped me in my research on sugarcane crops. I can keep track of field microclimatic conditions such as soil moisture, ambient temp. and humidity and take actions accordingly.


Astha
Islampur-Sangli
Maharashtra
Crops
Sugarcane



Mr. Vishwas Shinde 15 Acre

It was not possible to get good produce from farming while managing the job and daily chores. After installing SICCA Automation, management of fertigation and water has become possible in my farm and my mental and physical stress has reduced.


Shalgaon
Kadegaon - Sangli
Maharashtra
Crops
Sugarcane



Mr. Sagar Arve 60 Acre

Timely fertilizer and water management is very important for grape crop planning. With SICCA Automation, I can easily monitor my entire farm through a mobile app.


Boregaon
Tasgaon-Sangli
Maharashtra
Crops
Grapes



Adv. Kiran Uthle 10 Acre

Timely fertilizer and water management is necessary to reach the target of 100 tons of sugarcane per acre. This is not possible without strict planning of electricity, water and labour. I can achieve this planning using Sicca automation technology.


Yetgaon
Kadegaon - Sangli
Maharashtra
Crops
Sugarcane



Mr. Ganesh B. Shinde 20 Acre

Farming along with business is a difficult task. I had to spend a lot of time in farming and labor planning, so I chose SICCA Automation. This technology works 24 hours for me and takes care of all the planning of farming.


Kanheri
Baramati - Pune
Maharashtra
Crops
Papaya, Sugarcane

For more details: +91 800 712 6004 | mysicca.com

Thank you

Awards and recognition



MAHARASHTRA
STARTUP
WEEK 2020



Supported by MyGov & DeitY.
Anchored by CIIE, IIM-A



+91-9421058160

jasveer@wfmil.in

waterfallautomation.com





ANNEXURE



Smart Savings with SICCA automation



Saves upto*

80% Water

Schedule and manage irrigation activities from your mobile app and precisely irrigate using root-zone soil moisture sensors



Increases upto*

100% Yield

Precise and optimized irrigation and fertigation as per the crop need results in improved yield while also improving the immunity of the crop



Saves upto*

50% Fertilizers

Appropriate irrigation through SICCA system prevents nutrient leeching



Saves upto*

50% Time & Effort

Remotely control the farm's irrigation and fertigation from anywhere in a well-planned manner using a mobile phone without worry or physical hardships



Purandar pilot

Farmer name: Datta Shendkar
Period of experiment: Feb 2021 to June 2021



CROP

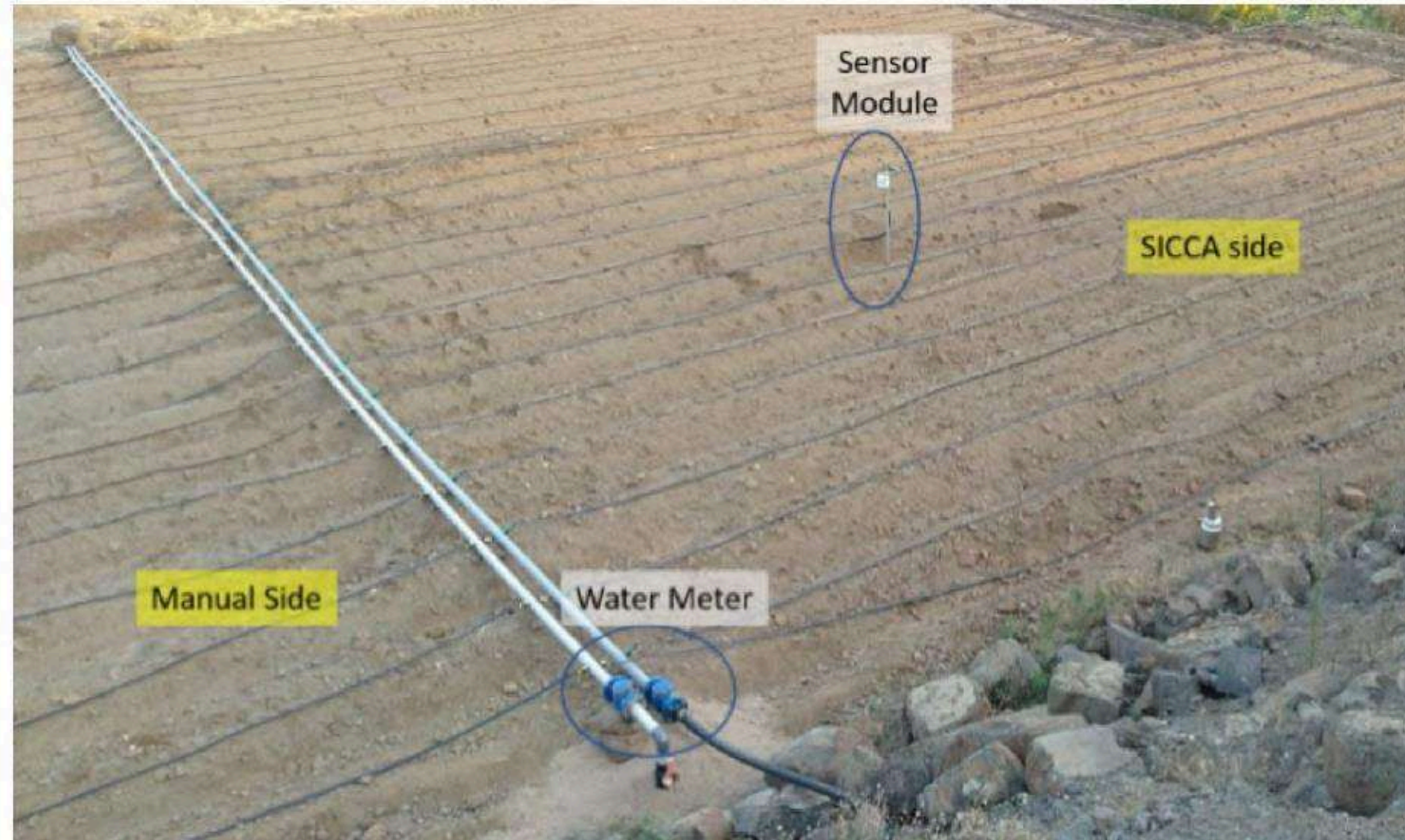
Marigold - *Kapri*



Total farm land owned: 7 acres
Crops cultivated: Flowers, Vegetables & Sugarcane
Village: Chambli | Taluka: Purandar
District: Pune Maharashtra, India

Experimental Setup

Land under experiment: 0.5 acres



- Farm split into 2 equal parts for parallel comparison experimentation – **Manual side & SICCA side**
- Each half had equal number of saplings installed.
- Both the halves had independent drip irrigation system with separate pumps.

Purandar pilot - Results



INCREASE IN YIELD:

SICCA side registered ~250 Kg of harvest & Manual side registered ~190 Kg of harvest. Thus, an increase in yield by ~30%.

Purandar pilot - Results (Contd.)



QUALITY OF PRODUCE:

Color retention on SICCA side & decolorization on Manual

Increase in Quality:

The decolorization of flowers on the Manual side fetched a price of Rs. 40 per Kg in the market while the good colored flowers on the SICCA side fetched Rs. 50 per Kg in flower market.

Thus, an increase in 25% of price.



WATER SAVINGS:

Final water meter reading with Manual Side showing 338.99 Kilolitres used. SICCA side showing 273.98 Kilolitres of water.

At the end of crop cycle, it was shown that Manual side used 65 Kilo-litres or 24% more water than the SICCA side i.e. 6-7 water tanks (water-saving of Rs. 6000 to 8000) in 4 months.



As a part of our mission to move communities towards sustainability, we are able make an impact over 9 SDGs